

**** This is the Analysis of Given GeoTiff (Multispectral Image) ****

Raster Information:

No. of Bands: 4

Width (pixels): 54

Height (pixels): 55

CRS: EPSG:32653

Transform:

| 3.00, 0.00, 636858.00|

| 0.00,-3.00, 4042014.00|

| 0.00, 0.00, 1.00|

Band Statistics:

Blue -> Min: 0, Max: 909, Mean: 287.88, Std: 280.11

Green -> Min: 0, Max: 1303, Mean: 553.07, Std: 518.66

Red -> Min: 0, Max: 1419, Mean: 470.51, Std: 464.32

NIR -> Min: 0, Max: 4513, Mean: 2087.12, Std: 1959.39

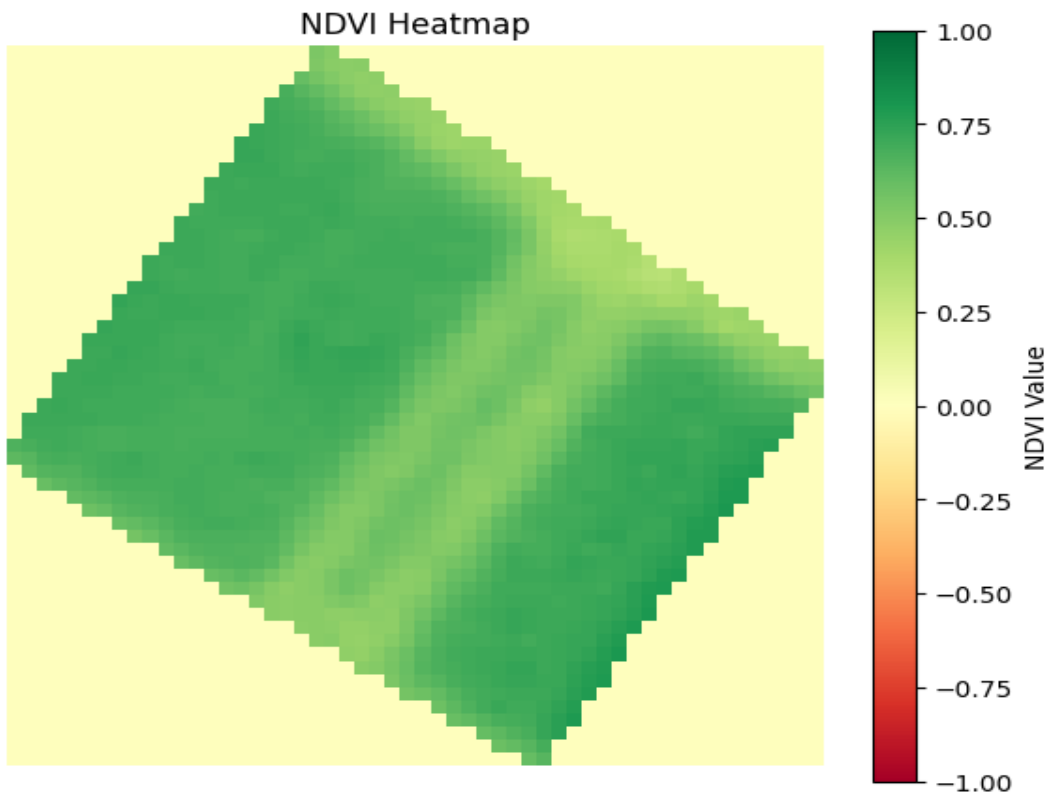
NDVI Statistics:

Min: 0.0000

Max: 0.7915

Mean: 0.3388

Std: 0.3221



NDVI Heatmap Analysis

NDVI Range :

-1 to 0 → Water bodies / Non-vegetated surfaces

0 to 0.2 → Bare soil / Built-up areas

0.2 to 0.4 → Sparse or stressed vegetation

0.4 to 0.6 → Moderately healthy vegetation

0.6 to 1.0 → Dense, healthy vegetation

Heatmap Observations

1. Dominant Colors: Light to Medium Green

- Indicates **moderate vegetation health**.
- Cropland or grassland condition.

2. Yellowish Patches

- Represent **sparse vegetation or bare soil**.

b. Could be areas with low plant cover.

3. Absence of Red Zones

a. No major **water bodies** or **completely barren land** visible.

NDWI statistics:

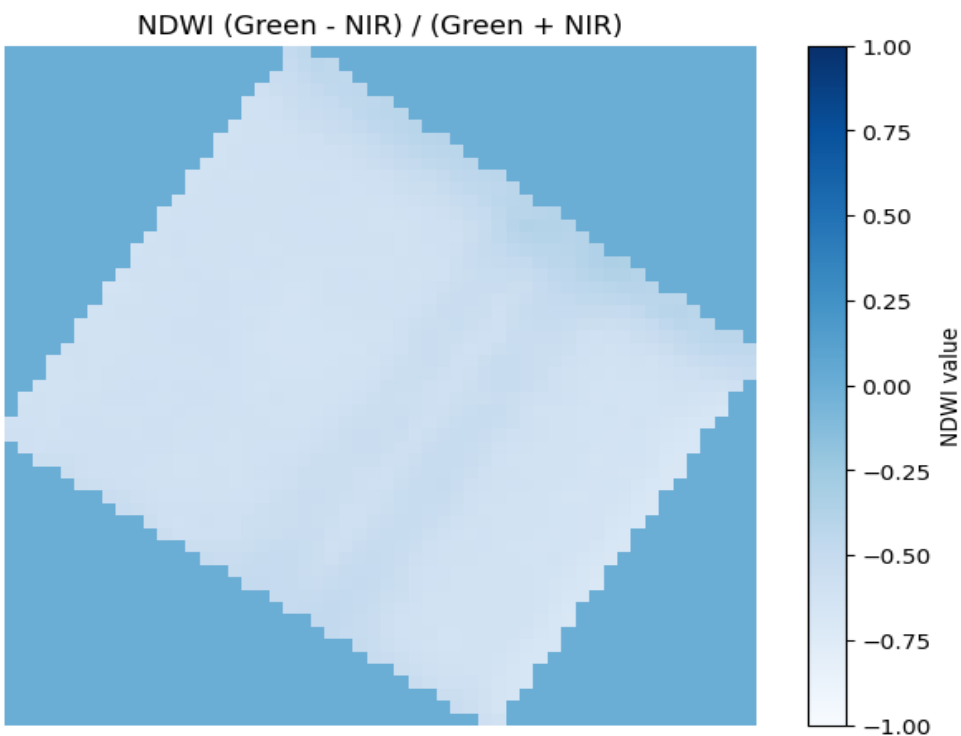
Min: -0.7066

Max: 0.0000

Mean: -0.3110

Std: 0.2913

NDWI Heatmap :



NDWI Heatmap Interpretation

- **Color Scale:**
 - **Dark Blue (~+1):** Indicates **high water content** (areas with open water bodies, ponds, rivers, or very wet soil).

- **Light Blue to White (~ 0):** Represents **moderate to low water content**, could be vegetation with some moisture or damp soil.
- **Pale Shades / Near 0 or Negative:** Indicates **dry areas**, bare soil, or built-up land with little to no water content.
- **Negative Values (< 0):** Typically correspond to **non-water surfaces** like soil, urban areas, or dry vegetation.

Created By group 2.....